The Modified Einstein–Planck Law with a Logarithmically Frequency-Dependent Quantum
From the localized-oscillator derivation to a self-consistent spectral law: demonstration, consistency checks, and parameter fit for the photon band (3 nm – 30 μm) Viorel-Mihai Popescu

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The Modified Einstein–Planck Law with a Logarithmically Frequency-Dependent Quantum

Abstract. The standard derivation of Planck's radiation law rests on a single postulate: the energy exchanged between a cavity mode of frequency ν and the radiation field is quantized in units of a *universal* constant, $\varepsilon = h\nu$. This document examines the minimal generalization in which the size of that quantum is allowed to vary weakly and logarithmically from mode to mode, $h \to h(\nu)$, while remaining independent of the occupation number n within a given mode. Part I shows, by direct inspection of the canonical-ensemble derivation, exactly which steps survive this generalization unchanged and which do not — this is the "justification/demonstration" in the sense of the companion document *From Localized Mechanical Oscillators to Planck's Radiation Law*. Part II introduces the additional, model-specific hypothesis needed to fix the functional form of $h(\nu)$: a self-consistent differential (rather than algebraic) relation between the spectral function and the photon quantum, motivated by a finite-volume ("time-quantum") picture of the photon gas. Part III fits the two free parameters of the resulting law to the requirement that photons at $\lambda_1 = 380.43$ nm and $\lambda_2 = 3.08$ nm carry the same energy, $E = 3.259$ eV. Part IV states, without qualification, the scope of what has and has not been demonstrated.

Part I — From the Canonical Oscillator to a Frequency-Dependent Quantum 1. Recap of the standard derivation

The standard result (derived in full in the companion note *From Localized Mechanical Oscillators to Planck's Radiation Law*) proceeds in four steps:

1. A localized mechanical oscillator of frequency ν obeys $\ddot{r} + 4\pi^2\nu^2 r = 0$, with conserved classical energy $E = 2\pi^2 m\nu^2 A^2$, continuous in the amplitude A.
2. Planck's postulate discretizes this energy: $E_n = n\,h\,\nu$, $n = 0,1,2,\ldots$, with h a **universal** constant, the same for every mode ν.

3. At temperature T, the canonical partition function of one mode is the geometric series
$$Z(\nu) = \sum_{n=0}^{\infty} e^{-nh\nu/k_BT} = \frac{1}{1 - e^{-h\nu/k_BT}},$$
giving the mean energy
$$\langle E(\nu)\rangle = \frac{h\nu}{e^{h\nu/k_BT} - 1}.$$
4. Multiplying by the mode density $g(\nu) = 8\pi\nu^2/c^3$ (a purely geometric count of standing waves in the cavity, independent of any assumption about h) gives Planck's law,
$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{e^{h\nu/k_BT} - 1}.$$

The only place where the *numerical value* of h enters this chain is step 2, and it enters as a single number, common to all modes.

2. The generalization: a mode-dependent, occupation-independent quantum

Hypothesis H1. Replace the universal constant h by a smooth function of frequency, $h(\nu)$, and postulate that the energy levels of a mode of frequency ν remain *equally spaced in* n, with spacing set by that mode's own value of h:
$$\boxed{E_n(\nu) = n\,\varepsilon(\nu), \qquad \varepsilon(\nu) \equiv \nu\,h(\nu), \qquad n = 0,1,2,\ldots}$$
This is the *only* structural change relative to the standard postulate: different modes may exchange energy in differently sized packets, but within one mode the packets are still all the same size (the level spacing does not depend on n).

2.1 Demonstration: which parts of the derivation survive

The claim to be verified is that steps 3–4 of §1 go through **unchanged in functional form**, with $h\nu$ replaced everywhere by $\varepsilon(\nu)$.

Partition function. Because $\varepsilon(\nu)$ does not depend on n, the sum over states of one mode is still a geometric series in $x \equiv e^{-\varepsilon(\nu)/k_BT}$:
$$Z(\nu) = \sum_{n=0}^{\infty} e^{-n\varepsilon(\nu)/k_BT} = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = \frac{1}{1 - e^{-\varepsilon(\nu)/k_BT}}.$$
Nothing in this sum used the fact that ε has the same value for every mode — only that it is constant *within* the sum over n at fixed ν. The convergence condition $x < 1$, i.e. $\varepsilon(\nu) > 0$, must hold at every ν, which restricts $h(\nu)$ to be positive on the domain of interest.

Mean energy. The two standard sums $\sum_n x^n = (1-x)^{-1}$ and $\sum_n n\,x^n = x(1-x)^{-2}$ (Appendix B) are algebraic identities in x; they do not reference $\varepsilon(\nu)$ at all except through the value of x. Hence
$$\langle E(\nu)\rangle = \varepsilon(\nu) \cdot \frac{\sum_n n\,x^n}{\sum_n x^n} = \frac{\varepsilon(\nu)}{e^{\varepsilon(\nu)/k_BT} - 1}.$$

Mean occupation number. Writing $\langle E(\nu)\rangle = \langle n(\nu)\rangle\,\varepsilon(\nu)$ identifies the Bose–Einstein occupation $\langle n(\nu)\rangle = \left(e^{\varepsilon(\nu)/k_BT} - 1\right)^{-1}$, exactly as in the standard case: **the Bose–Einstein statistics of the mode are unaffected by letting the quantum size vary from mode to mode.**

Mode density. $g(\nu) = 8\pi\nu^2/c^3$ is a statement about the geometry of standing waves in a cavity of volume V (§5 of the companion derivation); it does not involve h at all, so it is unaffected by Hypothesis H1.

Conclusion of the demonstration. Under H1 alone, the spectral energy density generalizes to
$$\boxed{u(\nu, T) = \frac{8\pi\nu^2}{c^3} \cdot \frac{\varepsilon(\nu)}{e^{\varepsilon(\nu)/k_BT} - 1}, \qquad \varepsilon(\nu) = \nu\,h(\nu)}$$
for **any** positive, smooth function $h(\nu)$ — this is the full content of allowing the Planck "constant" to run with frequency while keeping everything else about the oscillator/cavity picture untouched. Equation (2) is exact given H1; it fixes neither the form of $h(\nu)$ nor, by itself, any new physical content beyond a frequency-dependent substitution. Sections 4–9 introduce a second, independent hypothesis that *does* add new content, in the form of a self-consistency condition on $\varepsilon(\nu)$.

3. Choosing the functional form of $h(\nu)$

Equation (2) is compatible with an infinite family of functions $h(\nu)$. Three requirements narrow this family to a specific one-parameter ansatz:

1. **Reference-point normalization.** There must exist some frequency ν_0 at which the correction vanishes exactly, $h(\nu_0) = h_0$, so that the well-tested value of Planck's constant is recovered at that scale.
2. **Weak, perturbative deviation.** Over the frequency range of interest, $h(\nu)$ should differ from h_0 by a small, controlled amount, so that a first-order (linear) expansion in a small parameter ε_{dev} is meaningful.
3. **Scale covariance.** Because ν carries physical units, the only dimensionless combinations available to build a correction are ratios ν/ν_0 (or functions thereof). The simplest non-trivial, unbounded, monotonic dimensionless function of a ratio is its logarithm — the same structural argument used, by analogy, for

logarithmically "running" couplings in renormalization-group settings. A logarithm is also the unique choice consistent with requirement 1 without introducing a second free scale: $\ln(\nu/\nu_0) = 0$ automatically at $\nu = \nu_0$.

These three requirements select

$$\boxed{h(\nu) = h_0\left[1 + \varepsilon \ln\left(\frac{\nu}{\nu_0}\right)\right]}$$

with $|\varepsilon| \ll 1$ the deviation parameter and ν_0 the reference frequency at which $h(\nu_0) = h_0$ exactly. Substituting into $\varepsilon(\nu) = \nu h(\nu)$,

$$\boxed{E(\nu) \equiv \varepsilon(\nu) = h_0\,\nu\left[1 + \varepsilon \ln\left(\frac{\nu}{\nu_0}\right)\right]}$$

is the working ansatz analyzed for the remainder of this document. It is **not** itself derived from first principles — it is the simplest ansatz satisfying the three requirements above — but §2.1 shows rigorously that *whatever* smooth $h(\nu)$ is chosen, equation (2) for $u(\nu, T)$ follows without further assumption.

Part II — A Self-Consistent Differential Law: the Photon-Volume Model

Equation (2) already defines a complete, generalized spectral law once $h(\nu)$ is specified by (3). The purpose of this Part is different: it reproduces, in explicit step-by-step form, the additional physical model that was used (in the source notes *PlanckCalculus*) to *derive* — rather than merely assume — the specific parametric solution used in Part III. This model rests on a second hypothesis, independent of H1, and its status must be kept distinct: it is a structural postulate about how the photon quantum is generated, checked below (§8) for consistency with the $\varepsilon \to 0$ limit, not a theorem following from H1 alone.

4. Photon phase-space volume and the time quantum T_q

Model the photon gas occupying one mode as a finite volume, taken to be a cylinder of height $T_q\,c$ (light travels a distance $T_q c$ during the time quantum T_q) and circular cross-section of radius equal to the Bohr radius a_0:

$$V = T_q\,c\,\pi a_0^2.$$

Fixing the geometric closure condition $T_q\,c = a_0$ (the cylinder's height equals its base radius) turns (5) into $V = \pi a_0^3$ and identifies the time quantum as the light-crossing time of the Bohr radius:

$$T_q = \frac{a_0}{c} = \frac{5.29177 \times 10^{-11}\ \text{m}}{2.99792 \times 10^{8}\ \text{m/s}} = 1.765 \times 10^{-19}\ \text{s}.$$

T_q is the single new dimensional constant introduced by this model; it does not appear anywhere in Part I.

5. The master equation: statement and status

Hypothesis H2. Define the spectral function

$$F(\nu) = T_q \cdot 8\pi^2\nu^2\,\frac{E(\nu)}{e^{(E(\nu)+E_\mu)/E_0} - 1},$$

a frequency-weighted, exponentially damped analogue of $g(\nu)\langle E(\nu)\rangle$ built from the photon-volume constant T_q instead of $1/c^3$, and carrying two further constants: a minimum photon energy E_μ and a reference energy E_0.

Postulate that the photon energy is generated **self-consistently** as the frequency-derivative of its own spectral function:

$$\boxed{E(\nu) = A\,\frac{d}{d\nu}\left[\nu^2\,\frac{E(\nu)}{e^{(E(\nu)+E_\mu)/E_0} - 1}\right]} \qquad A \equiv 8\pi^2 T_q,$$

a first-order, implicit, nonlinear ordinary differential equation for $E(\nu)$.

Equation (8) is **not** a consequence of H1 or of the standard oscillator derivation; it is the model-specific postulate that fixes $E(\nu)$'s functional form beyond the ansatz (4). Its physical content is that the spectral function $F(\nu)$ is *generated by* the quantum it describes, rather than simply *containing* it multiplicatively as in the standard $u(\nu, T) = g(\nu)\langle E(\nu)\rangle$ — a differential rather than algebraic closure. The justification for adopting (8) is deferred to §8, where it is shown that the $\varepsilon \to 0$, $E_\mu \to 0$, $E_0 = k_B T$ limit of its solution has the same qualitative skeleton ($\nu^2 \times$ a function of $D_0 \equiv e^{h_0\nu/k_B T} - 1$) as the derivative of the standard Planck spectral density. This is a **consistency check**, not a derivation of (8) from (2): a differential closure of this kind is an additional structural choice, and the check below shows it does not contradict the $\varepsilon \to 0$ limit — it does not by itself prove that (8) is the unique or correct closure.

6. Explicit derivation of the nonlinear ODE

Introduce the auxiliary Bose-like weight

$$w(\nu) = \frac{1}{e^{(E(\nu)+E_\mu)/E_0} - 1}, \qquad D(\nu) \equiv e^{(E(\nu)+E_\mu)/E_0} - 1 = \frac{1}{w(\nu)},$$

and the inner function $f(\nu) = \nu^2 E(\nu)\,w(\nu)$, so that (8) reads $E = Af'$.

Product rule.

$$f'(\nu) = 2\nu E w + \nu^2(E'w + Ew').$$

Chain rule for w. Since $w = 1/D$ and $D = e^{(E+E_\mu)/E_0} - 1$,

$$\frac{dw}{d\nu} = -\frac{1}{D^2}\frac{dD}{d\nu} = -w^2 \cdot \frac{e^{(E+E_\mu)/E_0}}{E_0}\,E' = -w^2\,\frac{D+1}{E_0}\,E'.$$

Combine. Substituting (11) into (10),

$$f' = 2\nu E w + \nu^2 E' w - \nu^2 E w^2 \frac{D+1}{E_0} E'.$$

Using $E = Af'$ and collecting the terms proportional to E' on one side,

$$\frac{E}{A} - 2\nu E w = E' \left[\nu^2 w - \nu^2 E w^2 \frac{D+1}{E_0}\right],$$

which gives the **explicit first-order nonlinear ODE**

$$\boxed{E'(\nu) = \frac{E\left(\frac{1}{A} - 2\nu w\right)}{\nu^2 w\left(1 - E w \frac{D+1}{E_0}\right)}}$$

with $w = w(\nu)$ and $D = D(\nu)$ as in (9).

7. Perturbative expansion to first order in ε

Substitute the ansatz (4) and introduce dimensionless variables

$$x = \frac{h_0 \nu}{E_0}, \qquad \mu = \frac{E_\mu}{E_0}, \qquad \ell = \ln\left(\frac{\nu}{\nu_0}\right).$$

Let $D_0 \equiv e^{x+\mu} - 1$ denote the zeroth-order (constant-h) value of $D(\nu)$. Expanding $D(\nu)$ to first order in ε,

$$D(\nu) \approx D_0 + (D_0 + 1)\, x\, \varepsilon\, \ell.$$

The key algebraic step is the bracket that appears in the denominator of (12) once it is rewritten in terms of D:

$$D(\nu) - \frac{E(\nu)(D(\nu)+1)}{E_0}.$$

Expanding every factor to first order in ε and collecting terms,

$$D_0 - x(D_0 + 1) + \varepsilon\, \ell[x\,(D_0 + 1) - x\,(D_0 + 1)] = -1 + \mathcal{O}(\varepsilon^2).$$

The ε-dependent terms cancel identically at first order — this is not an approximation but an exact algebraic cancellation of the linear-in-ε coefficients, a direct consequence of $E(\nu)$ and $D(\nu)$ sharing the same $[1 + \varepsilon\ell]$ multiplicative structure at this order. As a result, (12) simplifies to

$$\frac{dF(\nu)}{d\nu} \approx \frac{8\pi^2 T_q\, \nu^2 h_0}{D_0^2}[(2D_0 - 1) + \varepsilon\, \ell\,(2D_0 - 1)] = T_q \cdot 8\pi^2\, \nu^2 h_0\, \frac{2D_0 - 1}{D_0^2}\left[1 + \varepsilon \ln\left(\frac{\nu}{\nu_0}\right)\right],$$

where $D_0 = \exp\left[(h_0\nu + E_\mu)/E_0\right] - 1$. Equation (16) is the first-order solution of the master ODE: it separates cleanly into a zeroth-order envelope, $\nu^2(2D_0 - 1)/D_0^2$, multiplying the same logarithmic factor already present in the ansatz (4).

8. Consistency checks: limiting cases

8.1 $\varepsilon \to 0$, arbitrary E_μ

$$\left.\frac{dF}{d\nu}\right|_{\varepsilon=0} = \frac{8\pi^2 T_q\, \nu^2 h_0\,(2D_0 - 1)}{D_0^2}, \qquad D_0 = e^{(h_0\nu + E_\mu)/E_0} - 1.$$

8.2 $\varepsilon \to 0$, $E_\mu \to 0$, $E_0 = k_B T$ — comparison with standard Planck

With these substitutions $D_0 = e^{h_0\nu/k_B T} - 1$, and the standard spectral density is

$$u_\nu = \frac{8\pi h_0 \nu^3}{c^3} \cdot \frac{1}{e^{h_0\nu/k_B T} - 1}.$$

Its frequency-derivative has the structure

$$\frac{d}{d\nu}\left[\frac{\nu^3}{e^{h_0\nu/k_B T} - 1}\right] = \frac{\nu^2\left[3D_0 + 3 - (D_0 + 1)\frac{h_0\nu}{k_B T}\right]}{D_0^2},$$

which shares with (17) the same overall skeleton $\nu^2 \times$ (function of $D_0)/D_0^2$, and reduces to the same leading power of D_0 in the regime $D_0 \gg 1$ (high frequency), where both (17) and (19) are dominated by their D_0-linear term. **This is the extent of the consistency check**: the master equation (8) does not contradict the standard theory in its relevant limit, but (17) and (19) are not algebraically identical term by term — equation (8) is a distinct closure relation, justified by this qualitative agreement and by the numerical fit of Part III, not by an exact reduction to (18).

9. Analytical solution via linearization at large ν

For large ν the ODE (8) is dominated by its leading balance. Assuming E' small (slowly varying E), (12) reduces to $E/A \approx 2\nu E w$, i.e. $w \approx 1/(2A\nu)$. Since $w \ll 1$ in this regime, $w \approx e^{-(E+E_\mu)/E_0}$, so

$$e^{(E+E_\mu)/E_0} \approx 1 + 2A\nu \quad \Rightarrow \quad E_{\text{eq}}(\nu) \approx E_0 \ln(1 + 2A\nu) - E_\mu.$$

This is the **leading-order (equilibrium) solution**. Writing a perturbed solution $E = E_{\text{eq}} + \delta(\nu)$ with $|\delta| \ll |E_{\text{eq}}|$ and substituting into (8), the leading balance cancels by construction of (20), and the remaining first-order terms in δ give the separable linear ODE

$$\delta'(\nu) \approx -\frac{2}{\nu}\,\delta(\nu) \quad \Rightarrow \quad \delta(\nu) = \frac{C}{\nu^2},$$

with C a constant of integration fixed by boundary/fit conditions. The **general linearized solution** is therefore

$$\boxed{E(\nu) = E_0 \ln(1 + 2A\nu) - E_\mu + \frac{C}{\nu^2}}$$

— the working formula used for parameter fitting in Part III.

Part III — Parameter Fit for the Photon Band (3 nm – 30 μm) 10. The equal-energy constraint

The model is required to reproduce, from equation (22), the same photon energy $E_T = 3.259$ eV at two target wavelengths:

$$\nu_1 = \frac{c}{\lambda_1} = \frac{2.99792 \times 10^8}{380.43 \times 10^{-9}} = 7.8804 \times 10^{14} \text{ Hz}, \qquad \nu_2 = \frac{c}{\lambda_2} = \frac{2.99792 \times 10^8}{3.08 \times 10^{-9}} = 9.7335 \times 10^{16} \text{ Hz}.$$

At the standard (unmodified) level, $h_0\nu_1 = 3.259$ eV exactly (by construction of λ_1), while $h_0\nu_2 = 402.55$ eV — more than two orders of magnitude away from the target. Three conditions fix the three free parameters A, E_0, C of (22):

Constraint I (recovery of the standard slope for small $2A\nu$):

$$2A\,E_0 = h_0.$$

Constraint II (equal energy at both target frequencies, with $E_\mu = 0$ in this version of the fit):

$$E_0 \ln(1 + 2A\nu_1) + \frac{C}{\nu_1^2} = E_T, \qquad E_0 \ln(1 + 2A\nu_2) + \frac{C}{\nu_2^2} = E_T.$$

Subtracting the two equations of (25) and substituting $E_0 = h_0/(2A)$ from (24) reduces the system to a single transcendental equation in A, whose solution is

$$\boxed{A = 4.2661 \times 10^{-15} \text{ s}, \qquad E_0 = 0.4847 \text{ eV}, \qquad C = 1.4085 \times 10^{30} \text{ eV Hz}^2}$$

Verification (direct substitution into (22)):

$$E(\nu_1) = 0.4847 \ln(1 + 2 \times 4.2661 \times 10^{-15} \times 7.8804 \times 10^{14}) + \frac{1.4085 \times 10^{30}}{(7.8804 \times 10^{14})^2} = 3.259 \text{ eV} \checkmark$$

$$E(\nu_2) = 0.4847 \ln(1 + 2 \times 4.2661 \times 10^{-15} \times 9.7335 \times 10^{16}) + \frac{1.4085 \times 10^{30}}{(9.7335 \times 10^{16})^2} = 3.259 \text{ eV} \checkmark$$

11. Equivalent logarithmic form of $h(\nu)$ and the value of ε

Setting the reference frequency $\nu_0 = \nu_1$ (so $h(\nu_0) = h_0$ at 380.43 nm), the condition $E(\nu_2) = E_T$ applied to the ansatz (4) gives

$$1 + \varepsilon \ln\left(\frac{\nu_2}{\nu_1}\right) = \frac{E_T}{h_0\nu_2} = \frac{3.259}{402.55} = 8.096 \times 10^{-3}.$$

Since $\ln(\nu_2/\nu_1) = \ln(123.52) = 4.8164$,

$$\boxed{\varepsilon = \frac{8.096 \times 10^{-3} - 1}{4.8164} = -0.20594}$$

so that

$$\boxed{h(\nu) = h_0 \left[1 - 0.20594 \ln\left(\frac{\nu}{7.8804 \times 10^{14} \text{ Hz}}\right)\right]}$$

$$\boxed{E(\nu) = h_0\,\nu \left[1 - 0.20594 \ln\left(\frac{\nu}{7.8804 \times 10^{14} \text{ Hz}}\right)\right]}$$

At ν_2, $h(\nu_2)/h_0 = 1/123.52 = 8.10 \times 10^{-3}$: the effective Planck constant is reduced to about 0.81% of its standard value at the soft-X-ray endpoint of the band. Note that $|\varepsilon| \approx 0.21$ is **not small** in an absolute sense; the "perturbative" label in §7 refers strictly to the algebraic expansion of the ODE, and its quantitative accuracy degrades toward the edges of the fitted band (§13).

12. Extended formula including the minimum photon energy E_μ

Re-fitting (22) with $E_\mu = -0.02007$ eV retained (loosely motivated by the muon-neutrino energy scale, but — see §16 — treated here purely as a free numerical constant) gives the alternative parametrization

$$\boxed{E(\nu) = -0.02007 + 151.5047 \ln(1 + 2.7872 \times 10^{-17}\,\nu) - 2.0632 \times 10^{-32}\,\nu^2 \quad \text{[eV]}}$$

This form and (30) are two independent numerical fits of the same underlying equation (22) to the same two-point constraint; they need not, and do not, coincide away from ν_1, ν_2.

13. Spectral energy table

Comparison of the standard energy $E_{\text{std}} = h_0\nu$ with the modified law (30), $E_{\text{mod}}(\nu) = h_0\nu[1 + \varepsilon \ln(\nu/\nu_0)]$, across the fitted band:

λ (nm)	Region	ν (Hz)	E_{std} (eV)	$h(\nu)/h_0$	E_{mod} (eV)	$\Delta E/E_{\text{std}}$
10 000	Far IR	2.998×10^{13}	0.1240	1.6732	0.2075	+67.3%
3 000	Mid IR	9.993×10^{13}	0.4133	1.4253	0.5890	+42.5%
1 000	Near IR	2.998×10^{14}	1.2398	1.1990	1.4866	+19.9%
700	Red	4.283×10^{14}	1.7712	1.1256	1.9936	+12.6%
550	Green	5.451×10^{14}	2.2543	1.0759	2.4254	+7.6%

λ (nm)	Region	ν (Hz)	E_{std} (eV)	$h(\nu)/h_0$	E_{mod} (eV)	$\Delta E/E_{\text{std}}$
450	Blue	6.662×10^{14}	2.7552	1.0346	2.8505	+3.5%
380.43	**UV ref**	7.880×10^{14}	**3.259**	**1.0000**	**3.259**	**0.0%**
300	UV	9.993×10^{14}	4.1328	0.9511	3.9306	−4.9%
200	Deep UV	1.499×10^{15}	6.1992	0.8676	5.3783	−13.2%
100	EUV	2.998×10^{15}	12.398	0.7248	8.987	−27.5%
30	Soft X-ray	9.993×10^{15}	41.328	0.4769	19.709	−52.3%
15	Soft X-ray*	2.000×10^{16}	82.656	0.3341	27.618	−66.6%
3.08	**Soft X-ray**	9.734×10^{16}	402.55	**0.00810**	**3.259**	**−99.19%**

*Below 30 nm the error of the first-order perturbative approximation (16) grows; below 15 nm it dominates and equation (30) should not be trusted quantitatively (§8.2, §16).

Part IV — Interpretation, Validity, and Scope 14. Why the same energy at two very different wavelengths?

From (30), $E(\nu_1) = E(\nu_2)$ requires

$$\frac{E(\nu_2)}{E(\nu_1)} = \frac{\nu_2}{\nu_1} \cdot \frac{1 + \varepsilon \ln(\nu_2/\nu_0)}{1 + \varepsilon \ln(\nu_1/\nu_0)} = 1.$$

Since $\nu_2/\nu_1 = 123.52$, the logarithmic factor must compensate this ratio almost exactly; with $\nu_0 = \nu_1$ (so $\ln(\nu_1/\nu_0) = 0$), the condition reduces to $h(\nu_2)/h_0 = 1/123.52$, already exhibited in §11. The two-wavelength coincidence is thus not a separate physical prediction but a restatement, by construction, of how ε was fixed.

15. Validity range and theoretical context

The fit is only intended to hold over $\lambda \in [30\text{ nm},\ 30\ \mu\text{m}]$, $\nu \in [10^{13},\ 10^{16}]$ Hz, with standard Planck behavior, $E \approx h_0\nu$, recovered near $\nu \approx \nu_0$. A logarithmically running $E(\nu) = h_0\nu[1 + \varepsilon \ln(\nu/\nu_0)]$ of the kind proposed here is structurally similar to phenomenological modified-dispersion-relation models (quantum-gravity phenomenology), to photon propagation in dispersive media, and to Lorentz-invariance-violation parametrizations — the analogy in §3 is offered in that spirit. It should be stressed that current laboratory bounds on any such frequency-dependence of h are extremely tight, $|\varepsilon| \lesssim 10^{-15}$, for tests of standard quantum electrodynamics; the value $|\varepsilon| \approx 0.21$ obtained here is **many orders of magnitude larger** than those bounds and is not offered as a correction competing with them. It is a parameter of this specific two-wavelength equal-energy model, not a proposed universal constant.

16. Scope, corrections applied, and explicit limitations

Relative to the original working notes (*PlanckCalculus*), two corrections were applied throughout this rewrite:

1. **Sign convention.** The sign of E_μ in $E(\nu) = E_0 \ln(1 + 2A\nu) - E_\mu + C/\nu^2$ is fixed consistently everywhere (equation (22) and its restatements), removing an alternation between $+E_\mu$ and $-E_\mu$ present in the original notes.
2. **Status of E_μ.** The original notes proposed several distinct physical identifications for E_μ across different spectral bands (muon-neutrino energy, "electronegativity of volume," minimum monopole energy, electron-neutrino energy), none of which matched the fitted numerical values in more than roughly half the cases examined. In this document E_μ is treated uniformly as a **free constant fixed by numerical fit only**, with no assumed correspondence to any named particle or process.

Finally, and most importantly: this document demonstrates the **internal mathematical consistency** of the hypothesis "h varies weakly and logarithmically with ν" — that its consequences can be derived step by step from an explicit generalization of the canonical oscillator picture (Part I) plus one additional, clearly flagged structural postulate (Part II, equation (8)), and that the resulting law can be fit self-consistently to the stated two-wavelength constraint (Part III). It does **not** demonstrate that this hypothesis is favored by, or even compatible with, the broad experimental record supporting a single universal Planck constant across all tested frequency ranges; that separate evaluation is the subject of other, dedicated analysis documents (see *Critical_Analysis_Modified_Planck_Law_v2* in this project).

Appendix A — Summary of the generalized derivation

Step	Result
Standard postulate (recalled)	$E_n = nh\nu$, h universal
H1 — mode-dependent quantum	$E_n(\nu) = n\,\varepsilon(\nu)$, $\varepsilon(\nu) = \nu h(\nu)$
Partition function (unchanged in form)	$Z(\nu) = (1 - e^{-\varepsilon(\nu)/k_B T})^{-1}$
Mean energy (unchanged in form)	$\langle E(\nu) \rangle = \varepsilon(\nu)/\left(e^{\varepsilon(\nu)/k_B T} - 1\right)$
Mode density (unaffected)	$g(\nu) = 8\pi\nu^2/c^3$
Log ansatz for $h(\nu)$	$h(\nu) = h_0[1 + \varepsilon \ln(\nu/\nu_0)]$
H2 — master ODE (new postulate)	$E = A\,\frac{d}{d\nu}\left[\nu^2 \frac{E}{e^{(E+E_\mu)/E_0} - 1}\right]$
First-order-in-ε solution	Eq. (16); ε-terms cancel to $\mathcal{O}(\varepsilon^2)$
Large-ν linearized solution	$E(\nu) = E_0 \ln(1 + 2A\nu) - E_\mu + C/\nu^2$
Fitted photon-band law	$E(\nu) = h_0\nu[1 - 0.20594 \ln(\nu/7.8804 \times 10^{14})]$

Appendix B — Mathematical identities used

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \qquad |x| < 1$$

$$\sum_{n=0}^{\infty} n\,x^n = \frac{x}{(1-x)^2}$$

$$\frac{d}{d\nu}\left[\frac{\nu^2 E(\nu)}{D(\nu)}\right] = \frac{1}{D^2}\left\{D \cdot \frac{d}{d\nu}[\nu^2 E(\nu)] - \nu^2 E(\nu)\,\frac{dD}{d\nu}\right\} \qquad \text{(quotient rule, §6)}$$

$$\frac{d}{d\nu}[\nu^2 E(\nu)] = 2\nu E(\nu) + \nu^2\,\frac{dE}{d\nu} \qquad \text{(product rule, §6)}$$

Appendix C — Summary of parameters

Parameter	Symbol	Value	Units	Meaning
Standard Planck constant	h_0	4.1357×10^{-15}	eV·s	Reference value
Deviation parameter	ε	-0.20594	—	Log slope of $h(\nu)$, eq. (28)
Reference frequency	ν_0	7.8804×10^{14}	Hz	$\lambda_0 = 380.43$ nm
Time quantum	T_q	1.765×10^{-19}	s	$= a_0/c$, eq. (6)
Log-ODE amplitude	A	4.2661×10^{-15}	s	$= h_0/(2E_0)$
ODE energy scale	E_0	0.4847	eV	$= h_0/(2A)$
Perturbation constant	C	1.4085×10^{30}	eV·Hz^2	From linearization, eq. (21)
Min. photon energy (free fit constant)	E_μ	-0.02007	eV	See §16 — no assumed particle identity
Target wavelength 1	λ_1	380.43	nm	UV–visible boundary
Target wavelength 2	λ_2	3.08	nm	Soft X-ray
Equal energy	E_T	3.259	eV	Constraint value
Validity range	λ	30 nm – 30 μm	—	Eq. (33)

References

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[2] M. Planck, "On the Law of the Energy Distribution in the Normal Spectrum," *Annalen der Physik* **4**, 553 (1901). Available: https://michaelvio.github.io/MPlanck.pdf

[3] A. Einstein, "On a Heuristic Viewpoint Concerning the Production and Transformation of Light" (1905). Available: https://einsteinpapers.press.princeton.edu/vol2-trans/100